

Lecture 12 Review Handout

Reproducibility | Statistics 367-1-4361

Lecture Overview

Lecture 12 steps back from individual statistical models and asks how scientific knowledge is produced, filtered, reported, and sometimes distorted. The lecture follows the scientific pipeline from study choice and design, through significance filtering, researcher degrees of freedom, questionable research practices, publication, incentives, and transparent reporting. The central message is that reproducible science requires more than correct code or a valid statistical model; it requires ownership of the full workflow. Bayesian workflow remains the statistical core, but this lecture extends the same logic of checking, transparency, and model criticism to the research process itself.

Lecture Sections

12.1 — Scientific knowledge

This section introduces reproducibility as a problem of scientific knowledge production, not only a problem of individual papers. The examples of chloroquine, MMR/autism, psychology, and cancer biology show that incorrect or fragile findings can enter the literature, influence behavior, and remain influential even after correction. The key idea is that scientific knowledge is the result of a long pipeline with many possible failure points.

New Definitions and Terms

- **Reproducibility crisis** — the recognition that many published findings are difficult to reproduce, replicate, or sustain in later work.
- **Retraction** — formal withdrawal of a paper from the scientific literature after serious problems are identified.
- **Replication** — a later study that tests whether an earlier empirical finding appears again under similar or related conditions.

Figure

Scientific knowledge is the result of a long pipeline



- Our course focused on one step
 - Plenty of possibilities for error in that step
- And also touched on data collection
 - And the possibility for errors in data collection
- Errors can arise in:
 - Study design
 - Publication
 - Knowledge dissemination

Figure 1: Slide 10 - scientific knowledge is the result of a long pipeline

12.2 — Where do studies come from?

This section rejects the simple textbook picture in which studies arise only from a clean logical sequence of hypothesis, experiment, and conclusion. Actual study choice is shaped by missing knowledge, available tools, funding priorities, collaborations, previous commitments, and what a group knows how to do. A good study design turns those pressures into a central yes/no question with a clear primary measure and a plausible expected effect.

New Definitions and Terms

- **Pilot data** — preliminary data used to develop methods, test feasibility, clarify analysis, support grants, or motivate new studies.

Figure

Does genetically modified food cause cancer?

- 200 rats for 2 years
 - Male and female
 - 10 groups
 - With different amounts and types of GM corn in the diet
- Measured
 - 34 organs
 - Blood samples and urine samples at 11 time points
 - 20 blood parameters
 - 16 urine parameters
 - Tumor incidence
 - Mortality
- Found kidney and liver toxicity starting at 90 days
 - And escalating to kidney and liver relate deaths

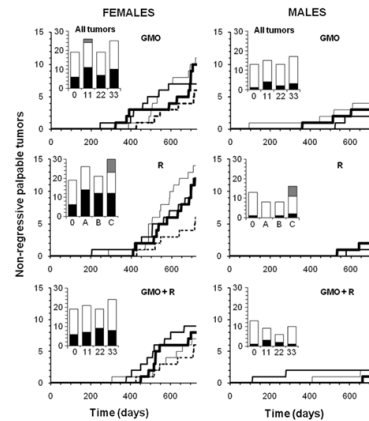


Figure 2: Slide 20 - criteria for good study design

12.3 — Fragile design

This section shows that problems can begin before analysis, when the design itself makes the result hard to interpret. The examples include low power combined with many measures, pseudo-replication, selection bias in observational studies, and confounding through experimenter expectations. The recurring lesson is that bad design is difficult to repair after the data have already been collected.

New Definitions and Terms

- **Pseudo-replication** — treating many measurements from the same experimental unit as if they were independent experimental units.
- **Selection bias** — systematic differences between groups caused by how participants or observations enter the study.
- **Confounding variable** — a variable that is mixed with the intended predictor or manipulation, making the causal interpretation unclear.

Figure

EFSA EFSA Journal 2012, doi:10.2903/j.efsa.2012.2986

Design is fragile: low power and a large number of measures

- Small number of rats per group
- Large number of measures
 - Called a 'fishing trip'
- Use of a strain that develops tumors with age naturally
- Inadequate statistical methods
- Lack of full reporting

- Paper originally retracted and then republished
 - But still derided

Figure 3: Slide 21 - design is fragile: low power and a large number of measures

12.4 — The significance filter

This section connects earlier material on significance to bias in the scientific literature. A threshold such as $p < 0.05$ turns continuous evidence into a binary distinction between “real” and “null” findings. Because only some results pass that threshold and many unpublished or non-significant results disappear, the published literature can overrepresent large, noisy, or lucky estimates.

New Definitions and Terms

- **Significance filter** — the process by which results are more likely to become visible when they cross a significance threshold.
- **File-drawer problem** — the disappearance of studies or analyses that do not produce publishable or significant results.
- **Multiple comparison testing** — correction for the fact that many tests create many chances for false-positive findings.

Figure

Turner *NEJM* 2008, doi:10.1056/NEJMsa065779

Failed drug trials do not get published

- Examined 74 submissions to FDA
 - Almost all the approved studies were published
 - Very few that were not approved were published

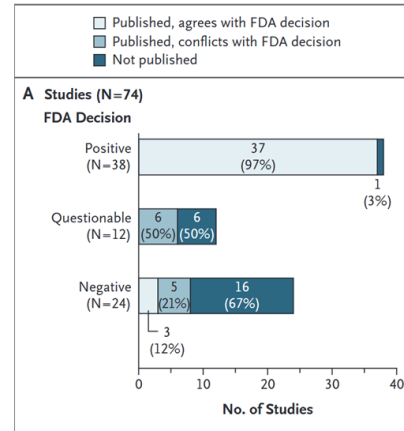


Figure 4: Slide 31 - selection biases which estimates get published

Eklund *PNAS* 2012, doi:10.1073/pnas.1602413113

Multiple tests multiply false positives

- An experiment can involve
 - Many voxels
 - Many electrodes
 - Many cells
 - Many hypotheses
- Correcting the significance is called multiple comparison testing
- Many methods
 - Bonferroni / Sidak
 - Random field theory
 - Permutation analysis
 - False discovery rate
- Different answers depending on the method chosen

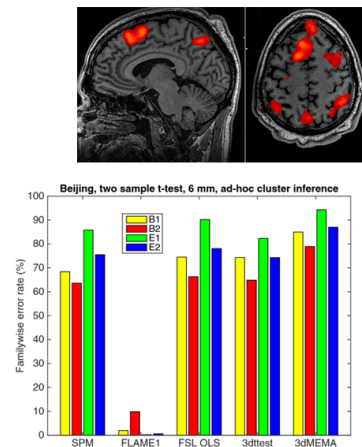


Figure 5: Slide 34 - multiple tests multiply false positives

12.5 — The garden of forking paths

This section moves from publication filtering to the many analysis choices researchers make before a result exists. Even without deliberate misconduct, different reasonable preprocessing choices, model choices, exclusion rules, and testing strategies can lead to different results. Specification curves and multiverse analyses make those unused paths visible rather than hiding them behind a single reported analysis.

New Definitions and Terms

- **Garden of forking paths** — the set of possible analysis decisions that could have been taken, often invisible in the final report.
- **Specification curve** — a display of results across many reasonable analytic specifications.
- **Multiverse analysis** — an analysis strategy that runs many plausible pipelines to examine how conclusions vary across choices.
- **Researcher degrees of freedom** — flexible decisions available to researchers that can change the result.

Figure

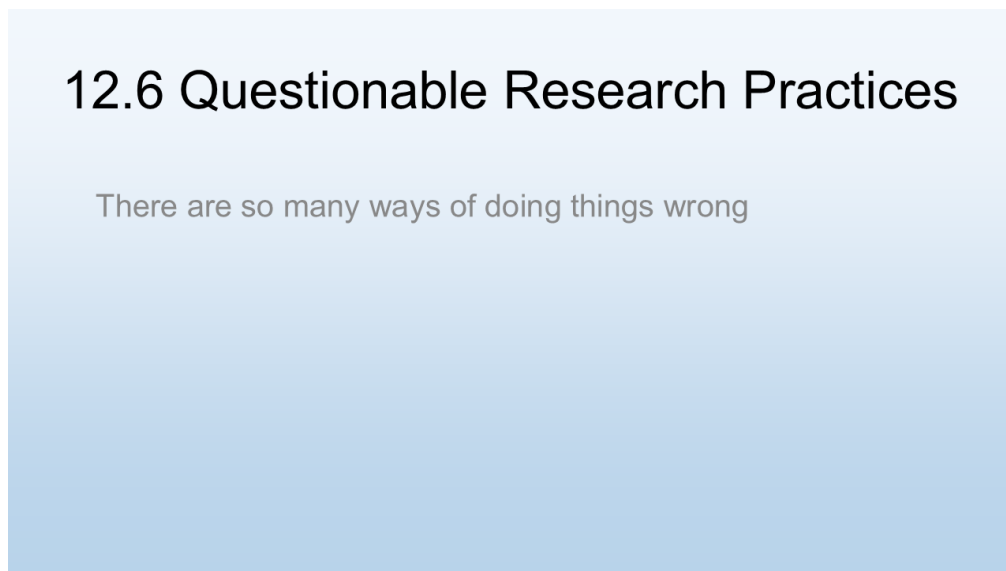


Figure 6: Slide 43 - specification curves can make forking paths visible

12.6 — Questionable Research Practices

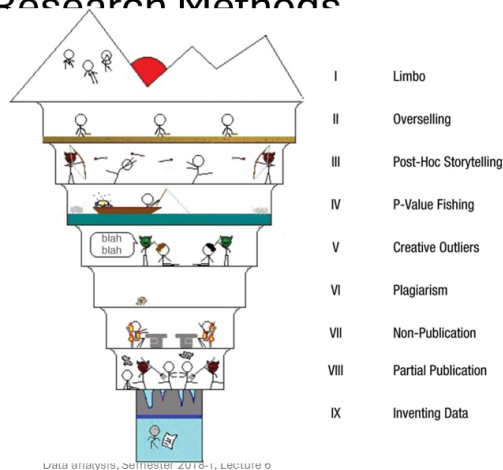
This section groups several failures under the heading of questionable research practices. The emphasis is that bad research can be unintentional, but some practices violate scientific values because they make the final claim look more confirmatory, cleaner, or more certain than it really is. p-hacking is highlighted as widespread, hard to detect, easy to do, and damaging; HARKing is treated as more ambiguous because storytelling and dishonesty can look superficially similar.

New Definitions and Terms

- **Questionable research practice (QRP)** — a research behavior that undermines scientific reliability without necessarily being outright fabrication or falsification.
- **p-hacking** — repeatedly changing analyses, exclusions, outcomes, or stopping rules until a significant result appears.
- **HARKing** — hypothesizing after the results are known; presenting a post-hoc explanation as if it had been the original hypothesis.

Figure

Questionable Research Methods



UCLA analysis, Semester 2018-1, Lecture 9

46

Figure 7: Slide 46 - central scientific values

Simulation of p-hacking and significance

Researcher degrees of freedom	Significance level		
	$p < .1$	$p < .05$	$p < .01$
Situation A: two dependent variables ($r = .50$)	17.8%	9.5%	2.2%
Situation B: addition of 10 more observations per cell	14.5%	7.7%	1.6%
Situation C: controlling for gender or interaction of gender with treatment	21.6%	11.7%	2.7%
Situation D: dropping (or not dropping) one of three conditions	23.2%	12.6%	2.8%
Combine Situations A and B	26.0%	14.4%	3.3%
Combine Situations A, B, and C	50.9%	30.9%	8.4%
Combine Situations A, B, C, and D	81.5%	60.7%	21.5%

Figure 8: Slide 49 - simulation of p-hacking and significance

12.7 — Exploration and Confirmation

This section rehabilitates exploration as legitimate scientific work while distinguishing it from confirmation. Exploratory science is appropriate when expectations, methods, or analyses are still being developed; confirmatory science is appropriate when the question, method, and story are already public and stable. Preregistration, Registered Reports, and transparent deviations are presented as tools for making the timing of decisions visible.

New Definitions and Terms

- **Preregistration** — recording hypotheses, methods, and analysis plans before observing the results.

Figure

Exploration, confirmation and harking

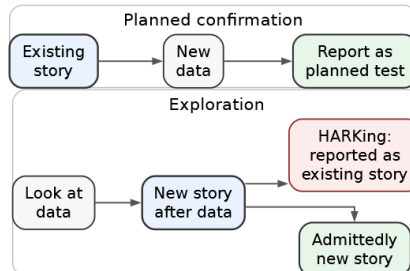


Figure 9: Slide 54 - exploration, confirmation and harking

Preregistration is a way of creating a timestamp

- Stage 1 review
 - Review science without knowing results
- Stage 2 review
 - Check that methods have been followed



Data analysis, 2020-2, Lecture 10

56/ 86

Figure 10: Slide 56 - preregistration is a way of creating a timestamp

12.8 — Creating the literature

This section asks how results become part of what other scientists read and use. Completion of a study is not the end of the process: results move through writing, submission, peer review, revision, and eventual acceptance or rejection. The literature includes primary papers, reviews, and meta-analyses, each with different roles in organizing, persuading, aggregating, or narrating scientific evidence.

New Definitions and Terms

- **Peer review** — evaluation of a submitted paper by external readers before publication.
- **Review paper** — a paper that organizes, interprets, argues about, or summarizes an area of research.
- **Meta-analysis** — a quantitative review that aggregates results, effect sizes, or significance patterns across studies.

Figure

A long path leads from results to acceptance



Figure 11: Slide 59 - a long path leads from results to acceptance

There are many forms of ‘literature’

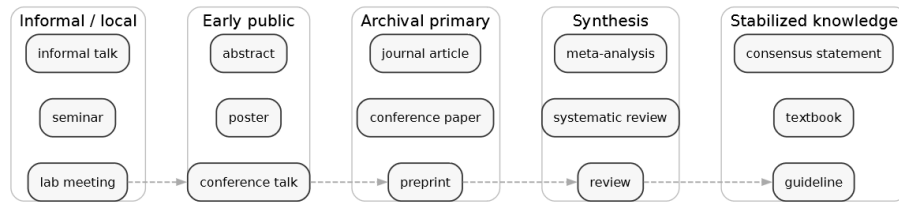


Figure 12: Slide 60 - there are many forms of literature

12.9 — Incentives and publication bias

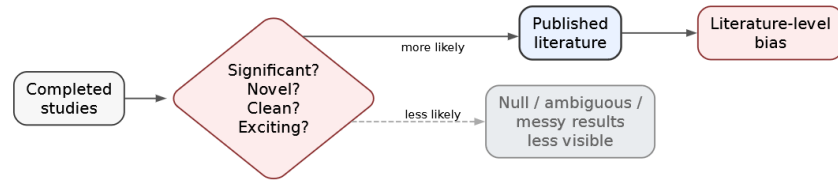
This section focuses on structural incentives rather than individual statistical choices. Positive results are easier to publish because they seem more interesting, fit narrative expectations, and are easier to interpret than failures to find an effect. The incentive system rewards publications, grants, hiring, promotion, and institutional visibility, so the literature is shaped by what the system rewards.

New Definitions and Terms

- **Publication bias** — systematic distortion of the literature because some results are more likely to be published than others.

Figure

It is always easier to publish positive results



- The literature needs to be interesting to scientists
- Our understanding needs a story
- Negative results can be poor technique
 - Harder to prove the lack of an effect

Figure 13: Slide 64 - it is always easier to publish positive results

Incentives for the scientist

- Everything depends on publications
 - Getting hired
 - Getting promoted
 - Getting funded
 - Getting students
 - Who can do experiments so that there can be more papers
- Institutions are also incentivized
 - Budgets
 - Special projects
 - Hiring quality faculty

Figure 14: Slide 66 - incentives for the scientist

12.10 — Transparent reporting

This section turns from diagnosing problems to reporting in a way that makes the literature better. Statistical reporting is needed in scientific papers, grant applications, preregistered research, ethical approval, medical reporting, engineering, and patents. The recommended structure separates what belongs in the main document, what belongs in an appendix, and what belongs in a repository.

New Definitions and Terms

- **Repository** — a public or shared location for data, code, additional analyses, and reproducible computational materials.

Figure

General reporting guidelines

- Three levels
 - Document
 - Methods
 - Results
 - Appendix
 - Model description
 - Diagnostics
 - Results
 - Repository
- Model description:**
- Model description with equations and diagram
 - Full specification of priors with statement on prior selection
 - Sampling algorithms and parameters of those algorithms
- Diagnostics:**
- Prior predictive plots and statistics
 - Diagnostic plots and statistics
 - Posterior predictive plots and statistics
- Results:**
- Table of parameters with
 - Mean
 - HDI
 - ESS and MCSE
 - Prior or prior HDI (if applicable)

72

Figure 15: Slide 72 - Transparent reporting guidelines for document, appendix, and repository

General reporting guidelines

- Three levels
 - Document
 - Methods
 - Results
 - Appendix
 - Model description
 - Diagnostics
 - Results
 - Repository
 - Data
 - Code
 - Additional analyses
- Data:
- Raw or preprocessed data should be in a public repository
- Code:
- A Python notebook or script that runs all relevant analyses
 - From raw data
 - Through modeling and diagnostics
 - And through posterior plotting and interpretation
 - To statistical inference and model comparison
 - Notebook should reproduce all figures and computations in the paper
- Additional analyses
- Analyses that did not make the paper
 - Additional models
 - Alternative prior

73

Figure 16: Slide 73 - Repository materials for data, code, and additional analyses

12.11 — Ownership of the workflow

The final section returns to the whole process. Understanding reproducibility requires taking responsibility not only for a final statistical model but also for study design, data collection, analysis choices, reporting, dissemination, and later interpretation. The complete workflow is therefore broader than the Bayesian workflow alone, but it preserves the same principle: make assumptions explicit, check each step, and keep the process transparent.

New Definitions and Terms

Figure

A complete workflow

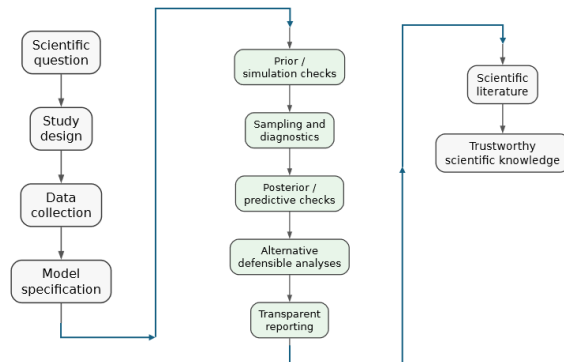


Figure 17: Slide 76 - a complete workflow